



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 1 • STANDARD 6.NS.3

Multiply Decimals

Lesson 1-6 • Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

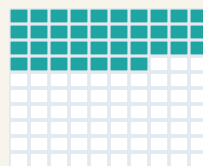
$$6 \times 4 = 24$$

Product = answer to multiplication

$4 \times 7 = 28$, so the product is 28

Product

Write the definition:



0.37

In 3.25, the decimal point separates 3 (ones) from 25 (tenths and hundredths)

Decimal point

Write the definition:

$$38 \times 21$$

≈

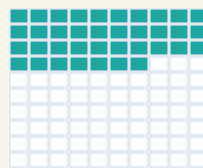
$$40 \times 20 = 800$$

about (rounded)

3.7×4.2 : round to $4 \times 4 = 16$, so the exact answer should be near 16

Estimate

Write the definition:

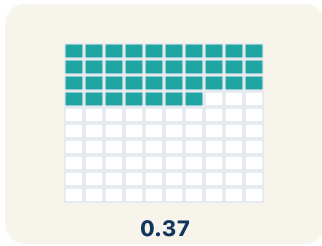


0.37

In 0.36: the 3 is in the tenths place (0.3) and the 6 is in the hundredths place (0.06)

Place value

Write the definition:



$$3.4 (1 \text{ place}) \times 2.6 (1 \text{ place}) = 8.84 (2 \text{ places total})$$

Decimal places

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can multiply decimals and place the decimal point correctly in the product.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Product

Decimal point

Estimate

Place value

Decimal places



Tap any word to see what it means and a picture.

1

The answer to a multiplication problem is the .

2

The dot that separates whole numbers from parts of a whole is the

.

3

Finding an answer that is close but not exact, to check if a product is reasonable, is to

.

4

The value a digit has based on its position is its .

5

To place the point in a product, I count the total number of

in both factors.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: What is 3.4×2.6 ?

- A. 8.84
- B. 88.4
- C. 0.884
- D. 8.48

Step 1 $34 \times 26 = 884$.

Step 2 There are 2 total decimal places ($1 + 1$), so place the decimal to get 8.84.

 **Answer:** A. 8.84

 Try — put the steps in order

Drag the cards (or use the ▲ ▼ buttons) to put the solution steps in the right order, then press **Check**.

There are 2 total decimal places ($1 + 1$), so place the decimal to get 8.84.

$34 \times 26 = 884$.

 We do — together

Solve this one with your class using the same steps.

Problem: What is 0.7×1.5 ?

- A. 1.05
- B. 10.5
- C. 0.105
- D. 1.50

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: What is 5.3×4 ?

- A. 21.2
- B. 2.12
- C. 212
- D. 20.12

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Each lemonade uses 0.6 of a cup of sugar, and sugar costs \$0.40 per cup. How much sugar cost is in one lemonade? (0.6×0.4)

- A. \$0.24
- B. \$2.40
- C. \$0.024
- D. \$24.00

Show your work:

2. The truck sells 2.5 pounds of fries, and each pound earns \$1.20 in profit. What is the total profit? (2.5×1.2)

- A. \$3.00
- B. \$30.00
- C. \$0.30
- D. \$3.70

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A family buys 3.5 pounds of deli meat at \$6.80 per pound. Show how to find the total cost. Use estimation to check if your answer is reasonable.

Sentence starter: The total cost is \$___ because $3.5 \times \$6.80 = \underline{\hspace{1cm}}$. I can estimate: $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$, which is close to my exact answer, so it is reasonable.

Show your work:

Reflect — Exit Ticket

What is 4.6×2.3 ?

- A. 10.58
- B. 105.8
- C. 1.058
- D. 10.48

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Product
2. **Notes 2:** Decimal point
3. **Notes 3:** Estimate
4. **Notes 4:** Place value
5. **Notes 5:** Decimal places
6. **We Try (notes):** A. $1.05 - 7 \times 15 = 105$. *There are 2 total decimal places (1 + 1), so place the decimal to get 1.05.*
7. **You Try (notes):** A. $21.2 - 53 \times 4 = 212$. *There is 1 total decimal place, so place the decimal to get 21.2.*
8. **Try It 1:** A. $\$0.24 - 6 \times 4 = 24$. *There are 2 total decimal places (1 + 1), so $0.6 \times 0.4 = 0.24$, which is \$0.24.*
9. **Try It 2:** A. $\$3.00 - 25 \times 12 = 300$. *There are 2 total decimal places (1 + 1), so $2.5 \times 1.2 = 3.00 = \3.00 .*
10. **Exit Ticket:** A. $10.58 - 46 \times 23 = 1,058$. *There are 2 total decimal places (1 + 1), so the answer is 10.58.*